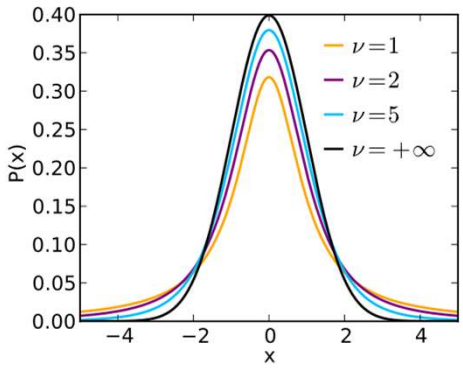
Agenda

- Last time
  - Central Limit Theorem
- Today
  - Estimates and Estimators
  - Means and proportions in the context of CLT
  - Standard Error and Approximations
  - The "student" distribution



1

Expectation and Variance of iid

- For identical X_i random variables, with $\mathbb{E}[X_i] = \mu$
  - $Y = X_1 + X_2 + \cdots X_n$ and $\bar{X} = Y/n$, $\mathbb{E}[\bar{X}] = \mu$
  - With independence, and finite variance σ^2, $\text{Var}(\bar{X}) = \sigma^2/n$
- Proportions
  - In the 1998 Teen survey, 1000 American teenagers (12 – 17 years) were interviewed. 27% reported having seen illegal drugs sold at school
  - Although this sort of thing is a hypergeometric distribution, we model this is W~ Bin(n, p) (why?)
    - Here W = Bin (1000, 0.27)
    - The random variable $\hat{P}$ = W/n represents proportions
  - $\mathbb{E}[\hat{P}] = \text{p}, \text{Var}(\hat{P}) = \text{p}(1-\text{p})/n$

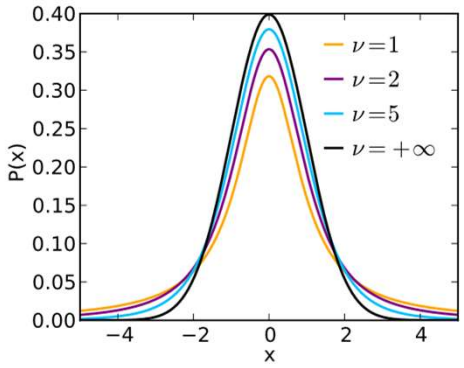
2

Parameter and estimates

- A ***parameter*** is a numerical characteristic of a population or distribution. It often is used to index the distribution
- You don't observe the whole population, you get a **random sample.**
  - A **statistic** is a random variable and a function of the sample random variables $X_1,...,X_n$ that does not depend on unknown parameters *before* observation
- **Estimator**: A statistic $T(X_1,...,X_n)$ that is specifically being used as a guess for an unknown parameter θ
- After you actually observe, an ***estimate*** is a realized quantity calculated from the data to estimate an unknown parameter
  - Capital letters refer to random variables
  - Small letters refer to observed values

3

CLT Approximation



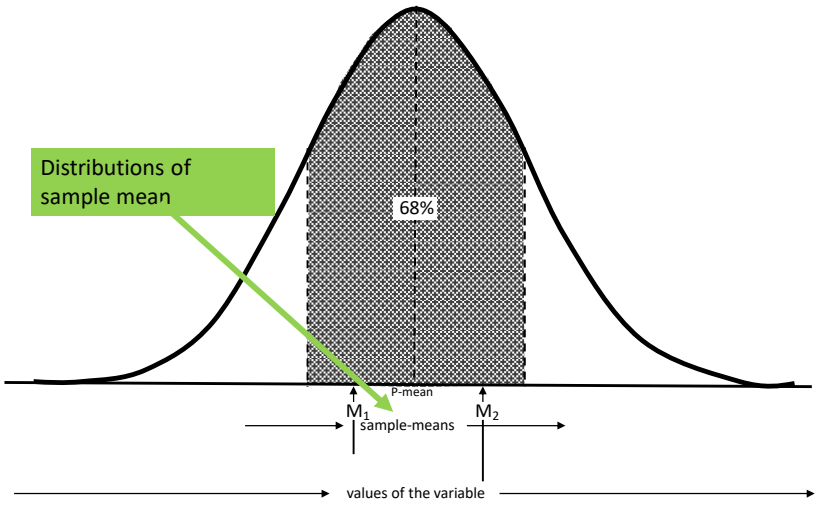
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CLT

- For independent identical (iid) random variables, and $Y = X_1 + X_2 + \cdots X_n$ with $\mathbb{E}[X_i] = \mu$ and finite variance σ^2
- Central limit theorem
  - For large (but finite) n, the distribution of Y is *approximately* normal with mean = $n\mu$ and variance $n\sigma^2$ (as if the distributions were normal)
  - The distribution of the sample mean $\overline{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$
  - Recall what we mean by "distribution of the sample mean"

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Distribution of sample means



6

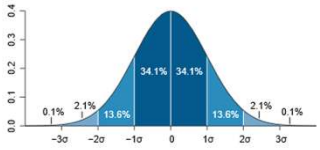
Example

- Observations $x_1, x_2, \ldots x_{100}$ [n=100] (or $x_1, x_2, \ldots x_{10}$) [n=10]
- Sample mean of observations: "estimate" denoted by $\overline{x}$
- If we want to ask questions like "what are the chances that the estimate will be …" we are abstracting and thinking about the random variable $\bar{X}$ (the estimator)
  - The CLT says that $\overline{X} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$ regardless of where X_i came from (for large n)

89	68	92	74	76	65	77	83	75	87
85	64	79	77	96	80	70	85	80	80
82	81	86	71	90	87	71	72	62	78
77	90	83	81	73	80	78	81	81	75
82	88	79	79	94	82	66	78	74	72

7

CLT Example



- Your friend tells you that in 10000 successive independent unbiased coin tosses, he counted at least 5200 heads. Is (s)he serious?
  - This is a single sample from your friend with choice of n=10000
- Let $X_1, X_2, \ldots, X_n$ be the random variables indicating whether the coin toss was a success (head).
  - These are i.i.d. random variables whose sum is a random variable with:
  - Expected value $n\mu = 10000(0.5) = 5000$, $\sigma = \sqrt{(10000 * 0.5 * 1.5)} = 50$
  - By CLT, $\bar{X} \approx \mathcal{N}(5000, 2500)$ and the z-score corresponding to 5200 (with continuity correction) is (5199.5-5000)/50 = 3.99 (4 standard deviations)
- Your friend is joking (why?)

Typo 0.5 not 1.5

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Review

- We use both $\bar{x}$ and $\bar{X}$ to refer to a sample mean. For what purposes do we use the former and for what purposes do we use the latter?
- What is meant by "the sampling distribution of $\bar{X}$?
- How is the population mean of the sample average $\bar{X}$ related to the population mean of individual observations?
- How is the population standard deviation of $\bar{X}$ related to the population standard deviation of individual observations?
- What happens to the sampling distribution of $\bar{X}$ if the sample size is increased?

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CLT for proportions

- 1000 American teenagers were interviewed. 27% reported having seen illegal drugs sold at school
- A single sample proportion is denoted by $\hat{p}$ (Here $\hat{p} = 0.27$)
- The corresponding random variable is $\hat{P}$
- The sample proportion $\hat{p}$ is an estimate of the true population parameter p from Bin(n,p)
- CLT: $\hat{P} \approx \mathcal{N}(p, \frac{p(1-p)}{n})$
  - In this example, B = Bin(1000, 0.27) itself is a (good) approximation, and the CLT is a further (good) approximation before

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CLT for proportions

- Your friend tells you that in 10000 successive independent unbiased coin tosses, he counted at least 5200 heads. Is (s)he serious?
- Let $\hat{p}$ be the sample proportion of heads
  - $E[\hat{p}] = p = 0.5,\ \ \mathrm{sd}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} = \sqrt{\frac{0.25}{10000}} = 0.005$
- With continuity correction, zscore $= \frac{\hat{p}-p}{\sqrt{p(1-p)/n}} = \frac{0.51995-0.5}{0.005} = 3.99$
  - Your friend is joking

Though not wrong, $se(\hat{p})$ is more appropriate

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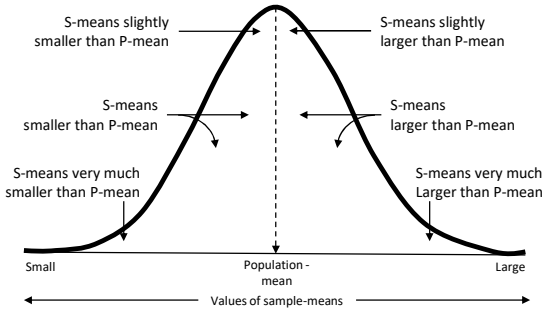
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Review

You want	Continuity correction on count	On proportion
$P(X \geq k)$	$k - 0.5$	$(k - 0.5)/n$
$P(X \leq k)$	$k + 0.5$	$(k + 0.5)/n$
$P(X = k)$	$[k - 0.5, k + 0.5]$	$[(k - 0.5)/n, (k + 0.5)/n]$

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Sample Mean vs Population Mean



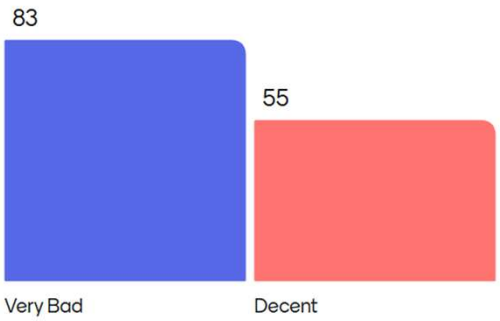
A single sample mean will give us a value that is different from the population mean

How bad is it?

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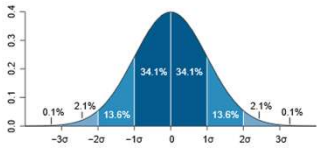
How bad is a single sample?

How bad is a single sample? **Can't say is not an option**



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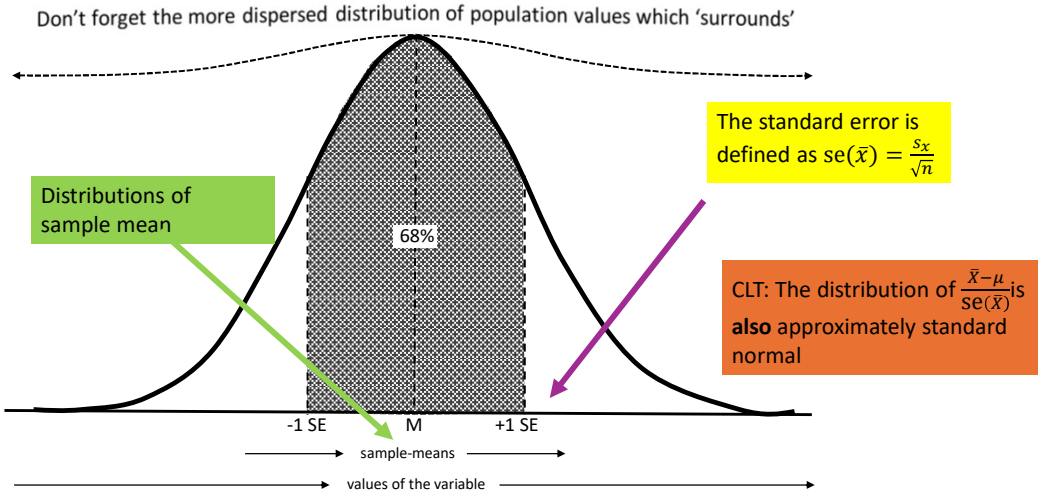
Logic of sampling



- The distance between $\bar{x}$ and μ is hardly bigger than $2\text{sd}(\bar{X})$ if we have a normal distribution for $\bar{X}$
  1. This could have been practically useless because nobody knows that their data is coming from a normal distribution
  2. No one knows the value of $\text{sd}(\bar{X})$
- However, CLT rescues us saying, yes, the population is normal in many cases, i.e., $\overline{\text{X}} \approx \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$ (note the approximation)
- We will address the second point shortly ("student distribution")
- Define: $\text{se}(\bar{x}) = \frac{s_x}{\sqrt{n}}$ where $s_x^2 = \frac{\sum_{i=1}^{n}(\text{x}_i - \overline{\text{x}})^2}{n-1}$
  - This is called the standard error of the sample mean

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Standard Error



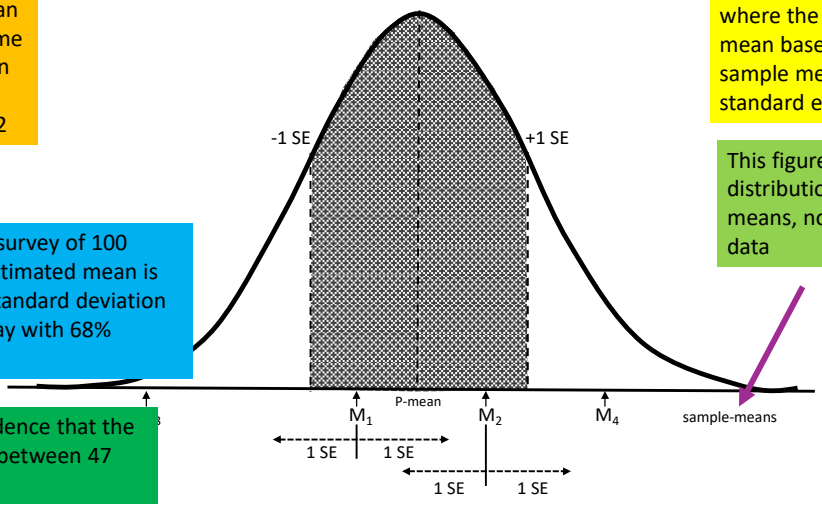
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Logic of Sampling

If the estimated mean is M2, 68% of the time our population mean will be within 1 standard error of M2

If we take a sample survey of 100 students, and the estimated mean is 50, and estimated standard deviation is 15, then we can say with 68% confidence that …

And with 95% confidence that the population mean is between 47 and 53



This figure tells us a very good idea about where the population mean based on a sample mean and standard error

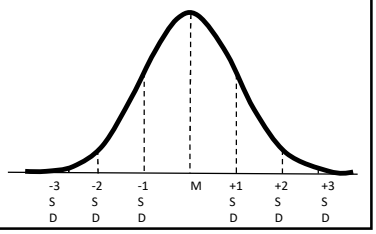
This figure is about the distribution of sample means, not the actual data

The population mean is fixed These are informal statements The exact def^ will appear in subsequent lectures The concept is illustrated here

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From Samples To Population

- Given a single pulse rate n=1, say 90, what can we say about the population distribution?
  - With n=1, our sample mean will not be accurate (but it won't be too off)
  - If we have a mean of a number of samples, we will have a sample standard deviation which will be different from the population standard deviation
- How wrong will we be? The size of the standard error of the mean depends on
  - The standard deviation within the sample
  - The size of the sample
  - The proportion of the population covered by the sample



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